Chapter 0: Matrix Algebra Review

Part I — Statistics Foundation for ML

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Abstract

This is chapter zero on purpose: everything from Chapter 2 onward (moment matrices, covariance matrices, multivariate distributions, and eventually regression coefficients and neural-network weight matrices) is written in matrix notation, and none of it is readable without this chapter first. This chapter reviews matrix multiplication, determinants, inverses, transposes, partitioned matrices, eigenvalues/eigenvectors, the four standard matrix decompositions (LU, Cholesky, QR, SVD), and matrix differentiation – the exact toolkit §4 (Matrix Notation for Moments) of Chapter 2 assumes you already have.

Acknowledgement: this chapter expands the STAT4ML prerequisite slides ("Matrix Calculus"), which drew on Barnes (2009) and Wikipedia's matrix decomposition article; see the reference list at the end.

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1 Matrix Multiplication

Definition 1.1 (Matrix product)**.** Let A be $m \times n$ and B be $n \times p$. The product $C = AB$ is the $m \times p$ matrix with entries

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad i = 1, \dots, m, \; j = 1, \dots, p.$$

Matrix multiplication is only defined when the inner dimensions match (A has as many columns as B has rows); the result takes the outer dimensions. This single rule is why a weight matrix in a neural network must have shape (output features $\times$ input features) – so it can left-multiply an input vector of the right length.

2 Determinants

For a 2×2 matrix $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $\det A = ad - bc$.

Theorem 2.1 (Properties of determinants)**.** For square matrices A, B of the same size:

1. $\det I = 1$.
2. Exchanging two rows of a matrix flips the sign of its determinant.
3. The determinant is a linear function of each row separately: for 2×2, $\begin{vmatrix} ta + a' & tb + b' \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$.
4. If two rows of A are equal, $\det A = 0$.
5. Subtracting t times row i from row j ($i \neq j$) does not change the determinant.
6. If A has an all-zero row, $\det A = 0$.
7. The determinant of a triangular matrix is the product of its diagonal entries.
8. $\det(AB) = (\det A)(\det B)$.
9. $\det(A^\top) = \det A$.

Try it yourself

Property 2 above is exactly why Gaussian elimination with row swaps has to track a sign flip when computing a determinant via row reduction – implement `det2x2` in this chapter's exercises and confirm property 8 numerically on two random 2×2 matrices.

3 The Inverse Matrix

Definition 3.1 (Invertible matrix)**.** An $n \times n$ matrix A is *invertible* if there exists an $n \times n$ matrix B with $AB = BA = I_n$. When it exists, B is unique, called the *inverse* of A, and denoted A^{-1}.

For a 2×2 matrix, $A^{-1} = \dfrac{1}{\det A} \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, which only exists when $\det A \neq 0$ – a matrix with

zero determinant is called *singular* and has no inverse. This is the same condition that shows up later as "$X^\top X$ must be invertible" for ordinary least squares to have a unique closed-form solution (Part II, Chapter 2).

4 Matrix Transpose

Definition 4.1 (Transpose)**.** The transpose of A, written $A^\top$, swaps A's row and column indices: $(A^\top)_{ij} = A_{ji}$.

Example 4.1. For $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{smallmatrix}\right)$, $A^\top = \left(\begin{smallmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{smallmatrix}\right)$.

Theorem 4.1 (Properties of transpose and inverse)**.**

1. $(A^\top)^\top = A$.
2. $(A \pm B)^\top = A^\top \pm B^\top$.
3. $(\kappa A)^\top = \kappa A^\top$ for a scalar κ.
4. $(AB)^\top = B^\top A^\top$ (order reverses!).
5. If $C = AB$ with A, B both invertible, $C^{-1} = B^{-1}A^{-1}$ (order reverses here too).

5 Partitioned Matrices

Definition 5.1. Write A $(m \times n)$ in block form $A = \left(\begin{smallmatrix} B & C \\ D & E \end{smallmatrix}\right)$, where B is $m_1 \times n_1$, E is $m_2 \times n_2$, C is $m_1 \times n_2$, D is $m_2 \times n_1$, $m_1 + m_2 = m$, $n_1 + n_2 = n$. This is a *partition* of A.

Partitioned matrices are exactly the shape of the block covariance matrix $\mathrm{Cov}\left(\begin{smallmatrix} Y \\ Z \end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathrm{Cov}(Y) & \mathrm{Cov}(Y,Z) \\ \mathrm{Cov}(Z,Y) & \mathrm{Cov}(Z) \end{smallmatrix}\right)$ from Chapter 2, §4 – this chapter's block-matrix notation and that chapter's block covariance notation are literally the same object.

6 Eigenvalues and Eigenvectors

Almost every vector changes direction when multiplied by a matrix A. The exceptional vectors that do not – that come back out parallel to themselves – are **eigenvectors**.

Definition 6.1 (Eigenvalue, eigenvector)**.** For a square matrix A, a nonzero vector x is an *eigenvector* of A with *eigenvalue* λ if

$$Ax = \lambda x.$$

λ tells you whether x is stretched, shrunk, reversed, or left unchanged by A.

Example 6.1. Let $A = \left(\begin{smallmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{smallmatrix}\right)$. Solving $\det(A - \lambda I) = 0$:

$$\det\left(\begin{smallmatrix} 0.8-\lambda & 0.3 \\ 0.2 & 0.7-\lambda \end{smallmatrix}\right) = \lambda^2 - \tfrac{3}{2}\lambda + \tfrac{1}{2} = (\lambda - 1)(\lambda - \tfrac{1}{2}) = 0,$$

so $\lambda_1 = 1$, $\lambda_2 = 1/2$.

Theorem 6.1 (Trace and determinant, 2×2)**.** For a 2×2 matrix A with eigenvalues λ_1, λ_2 (possibly complex, possibly repeated):

$$\lambda_1 + \lambda_2 = \operatorname{trace}(A) = a_{11} + a_{22}, \qquad \lambda_1\lambda_2 = \det A.$$

This gives a fast sanity check for any eigenvalue computation you write: the eigenvalues must sum to the trace and multiply to the determinant – Exercise 3 in this chapter builds exactly this check into its test suite.

7 Matrix Decompositions

A **matrix decomposition** factors a matrix into a product of (usually simpler, structured) matrices. Decompositions related to solving linear systems: LU, Cholesky, QR. A decomposition based on eigenvalues: SVD.

7.1 LU Decomposition

Definition 7.1. The *LU decomposition* writes a square matrix as $A = LU$, with L lower triangular and U upper triangular.

Example 7.1. For $A = \left(\begin{smallmatrix} 4 & 3 \\ 6 & 3 \end{smallmatrix}\right)$, write $A = LU$ with $L = \left(\begin{smallmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{smallmatrix}\right)$, $U = \left(\begin{smallmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{smallmatrix}\right)$. Fixing $\ell_{11} = \ell_{22} = 1$ (Doolittle's convention) and expanding LU entrywise gives four equations:

$$\ell_{11}u_{11} = 4, \quad \ell_{11}u_{12} = 3, \quad \ell_{21}u_{11} = 6, \quad \ell_{21}u_{12} + \ell_{22}u_{22} = 3.$$

Solving in order: $u_{11} = 4$, $u_{12} = 3$, $\ell_{21} = 6/4 = 1.5$, and $\ell_{22}u_{22} = 3 - 1.5(3) = -1.5 \Rightarrow u_{22} = -1.5$. So

$$\left(\begin{smallmatrix} 4 & 3 \\ 6 & 3 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 1.5 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 4 & 3 \\ 0 & -1.5 \end{smallmatrix}\right).$$

Editorial note

The original prerequisite slide this example is drawn from has a digit typo at exactly this step – it states "$u_{11} = 14$" where it means $u_{11} = 4$ (everything downstream in the original slide is computed correctly using $u_{11} = 4$, so it is an isolated typo, not a repeated error). See `AUDIT.md` at the repo root.

7.2 Cholesky Decomposition

For a Hermitian positive-definite matrix A, the *Cholesky decomposition* is $A = LL^{\top}$ with L lower triangular and positive diagonal entries:

$$L_{jj} = \sqrt{A_{jj} - \sum_{k=1}^{j-1} L_{jk}^2}, \qquad L_{ij} = \frac{1}{L_{jj}}\left(A_{ij} - \sum_{k=1}^{j-1} L_{ik}L_{jk}\right) \text{ for } i > j.$$

Because it needs only one triangular factor (not two), Cholesky is the standard way to solve $\Sigma x = b$ when Σ is a covariance matrix (always symmetric positive-(semi)definite) – about twice as fast as a generic LU solve.

7.3 QR Decomposition

$A = QR$, with Q orthogonal ($Q^\top Q = I$) and R upper triangular. One construction method is Gram-Schmidt: given columns $a_1, \dots, a_n$ of A, set $u_1 = a_1$, $e_1 = u_1/\|u_1\|$, and iteratively

$$u_{k+1} = a_{k+1} - \sum_{j=1}^{k} (a_{k+1} \cdot e_j) e_j, \qquad e_{k+1} = \frac{u_{k+1}}{\|u_{k+1}\|}.$$

Then $Q = [e_1 | \cdots | e_n]$ and $R_{ij} = a_j \cdot e_i$ for $i \le j$ (0 below the diagonal).

7.4 Singular Value Decomposition (SVD)

The SVD generalizes eigendecomposition to *any* $n \times p$ matrix (not just square ones):

$$A_{n\times p} = U_{n\times n}\, S_{n\times p}\, V^\top_{p\times p},$$

with $U^\top U = I_n$, $V^\top V = I_p$ (both orthogonal). The columns of V are the eigenvectors of $A^\top A$; the columns of U are the eigenvectors of $AA^\top$; the singular values in S are the square roots of the (shared, nonnegative) eigenvalues of $A^\top A$ or $AA^\top$. SVD is what PCA (Part II, Chapter 5) actually computes under the hood.

8 Matrix Differentiation

Definition 8.1 (Jacobian)**.** Let $y = \psi(x)$ with y an m-vector and x an n-vector. The *Jacobian* $\dfrac{\partial y}{\partial x}$ is the $m \times n$ matrix with (i, j) entry $\dfrac{\partial y_i}{\partial x_j}$.

A handful of identities cover almost every derivative you need in this course (all assuming A does not depend on the variable being differentiated with respect to, unless stated otherwise):

$$\begin{aligned}
y = Ax \;&\Rightarrow\; \frac{\partial y}{\partial x} = A. \\
\alpha = y^\top A x \;&\Rightarrow\; \frac{\partial \alpha}{\partial x} = y^\top A, \qquad \frac{\partial \alpha}{\partial y} = x^\top A^\top. \\
\alpha = x^\top A x \ (A \text{ not symmetric}) \;&\Rightarrow\; \frac{\partial \alpha}{\partial x} = x^\top (A + A^\top). \\
\alpha = x^\top A x \ (A \text{ symmetric}) \;&\Rightarrow\; \frac{\partial \alpha}{\partial x} = 2x^\top A. \\
\alpha = x^\top x \;&\Rightarrow\; \frac{\partial \alpha}{\partial x} = 2x^\top.
\end{aligned}$$

The symmetric quadratic-form identity, $\partial(x^\top A x)/\partial x = 2x^\top A$, is the one you will use over and over: it is exactly the gradient of the ridge-regression penalty $\lambda \beta^\top \beta$ (Part II, Chapter 3) and of the Gaussian log-density's quadratic term.

Theorem 8.1 (Derivative of a matrix inverse)**.** Let $A(\alpha)$ be a nonsingular $m \times m$ matrix whose entries are differentiable functions of a scalar α. Then

$$\frac{\partial A^{-1}}{\partial \alpha} = -A^{-1} \frac{\partial A}{\partial \alpha} A^{-1}.$$

Editorial note

The original prerequisite slide states this identity *without* the leading minus sign (as $A^{-1}\frac{\partial A}{\partial \alpha}A^{-1}$). That is incorrect – differentiate $AA^{-1} = I$ with respect to α using the product rule to see why the minus sign is required: $\frac{\partial A}{\partial \alpha}A^{-1} + A\frac{\partial A^{-1}}{\partial \alpha} = 0 \Rightarrow \frac{\partial A^{-1}}{\partial \alpha} = -A^{-1}\frac{\partial A}{\partial \alpha}A^{-1}$. See `AUDIT.md`.

9 Summary

Summary

- Matrix multiplication requires matching inner dimensions; $(AB)^\top = B^\top A^\top$ and $(AB)^{-1} = B^{-1}A^{-1}$ both reverse order.
- A matrix is invertible iff its determinant is nonzero; for 2×2, $A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
- Eigenvalues/eigenvectors satisfy $Ax = \lambda x$; for 2×2, eigenvalues sum to the trace and multiply to the determinant – a fast correctness check for eigenvalue code.
- LU, Cholesky, and QR decompose a matrix for solving linear systems; SVD generalizes eigendecomposition to non-square matrices and underlies PCA.
- $\partial(x^\top Ax)/\partial x = 2x^\top A$ for symmetric A is the single most-reused matrix-calculus identity in this course.

10 Exercises

Three exercises per language (`exercises/{python,r,rust}/`):

1. **Exercise 1 – Core operations:** matrix multiply, transpose, 2×2 determinant and inverse, verifying $(AB)^\top = B^\top A^\top$ and $\det(AB) = \det(A)\det(B)$.
2. **Exercise 2 – LU decomposition:** Doolittle's method for a 2×2 matrix, verified by reconstructing A from L and U.
3. **Exercise 3 – Eigenvalues/eigenvectors:** the 2×2 characteristic-polynomial formula, verified against the trace/determinant identities above.

Each has a Rustlings-style broken counterpart in `exerciseNN_buggy.{py,R,rs}`.

References

- Barnes, R. J. (2009). *Matrix Differentiation (and some other stuff).*
- Wikipedia contributors. *Matrix decomposition.* Wikipedia, The Free Encyclopedia.